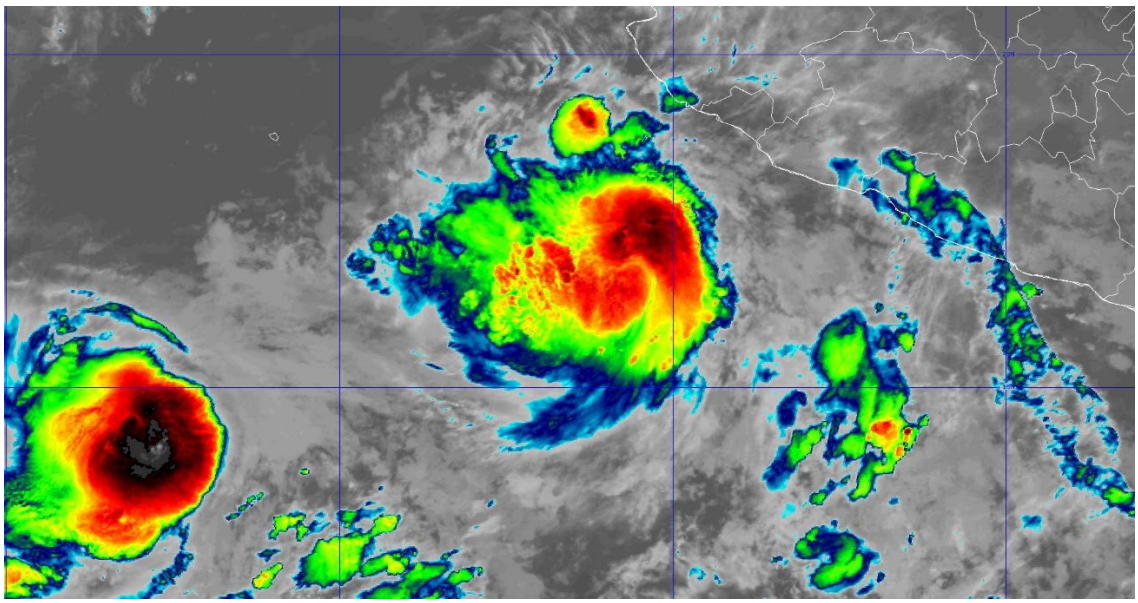
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

HURRICANE BARBARA

(EP022025)

8–10 June 2025

Andrew B. Hagen
National Hurricane Center
31 July 2025



GOES-WEST INFRARED SATELLITE IMAGE OF HURRICANE BARBARA AT 1100 UTC 9 JUNE 2025, NEAR THE TIME OF ITS PEAK INTENSITY. TROPICAL STORM COSME APPEARS IN THE LOWER LEFT CORNER OF THE IMAGE. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Barbara was a category 1 hurricane (on the Saffir-Simpson Hurricane Wind Scale) that remained offshore of southwestern Mexico.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Barbara.

Hurricane Barbara

8–10 June 2025

BEST TRACK

The "best track[2]" positions and intensities for Hurricane Barbara are listed in Table 1. The "best track" chart of Barbara's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

One ship reported tropical-storm-force winds in association with Barbara. The ship with call sign 9V6864 measured 35-kt winds in Barbara's northeast quadrant on 8 June at 1800 UTC.

Origin

Barbara originated from a disturbance along the eastern North Pacific monsoon trough a few hundred miles south of the Gulf of Tehuantepec on 4 June. A tropical wave that emerged off the west coast of Africa on 24 May and crossed Central America on 2–3 June may have also played a role in the genesis of Barbara. The disturbance began moving west-northwestward on 5 June. Shower and thunderstorm activity started to become better organized on 7 June, and it is estimated that a tropical storm formed around 0600 UTC 8 June while centered roughly 150 n mi offshore the coast of the Mexican state of Guerrero.

Peak Intensity and Minimum Pressure

Barbara's peak intensity of 65 kt from 0600 UTC to 1800 UTC 9 June is based on subjective Dvorak estimates as well as UW-CIMSS objective intensity estimates. TAFB and SAB estimates both peaked at T4.0/65 kt on 9 June. In addition, the SATCON, DPRINT and DMINT algorithms all supported hurricane intensity that day, peaking in the 65 to 70 kt range.

The estimated minimum central pressure of 991 mb is based on the Knaff-Zehr-Courtney (KZC) pressure-wind relationship.

---

[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Barbara.

CASUALTY AND DAMAGE STATISTICS

There were no reports of significant damage or casualties associated with Barbara.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The genesis area was introduced in the 7-day low category well in advance of formation (over 9 days). The 7-day probability was raised to the medium category nearly 8 days prior to formation and to the high category 108 h in advance. The 2-day genesis forecasts provided sufficient lead time, reaching the medium category 60 h in advance and the high category 30 h prior to formation. Figure 4 shows composites of 7-day TWO genesis areas for each category prior to the formation of Barbara. The actual genesis location fell outside and to the west-northwest of the forecast genesis ellipse areas for several of the forecasts made 7 to 9 days before genesis, because genesis occurred later than expected. Overall, 74 percent of the genesis forecasts correctly encompassed the genesis location. Of the genesis forecasts that were issued within 7 days of genesis, 97 percent correctly encompassed the genesis location.

A verification of NHC official track forecasts for Barbara is given in Table 3a. Official track forecast errors were lower than the mean official errors for the previous 5-yr period at 12–36 h, but higher than the 5-yr mean at 48 h, albeit for a small sample size. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. Of note is the poor performance of the regional hurricane models HFAI, HFBI, and HWFI. The poor performance appears to be because those models showed that Tropical Storm Cosme, located southwest of Barbara (cover photo), would have a significant influence on Barbara's track during the weakening phase, with Barbara moving westward to the north of Cosme, within Cosme's more dominant circulation (Fig. 5). In actuality, Cosme had much less of an impact on Barbara's track, and Barbara moved northward and northeastward after it weakened to a low-end tropical storm (Fig. 6). The aforementioned regional models were not the only models that initially indicated that Barbara would have more interaction with Cosme, and this is shown in Table 3b with the relatively large track errors by many of the models at 36 and 48 h.

A verification of NHC official intensity forecasts for Barbara is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period, except at 36 h. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b.

There were no coastal watches or warnings issued for Barbara.

ACKNOWLEDGEMENTS

Dr. Philippe Papin (NHC) created the Graphical TWO verification figure.

Table 1. Best track for Hurricane Barbara, 8–10 June 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
08 / 0600	15.0	101.9	1002	40	tropical storm
08 / 1200	15.4	102.7	998	50	"
08 / 1800	15.8	103.7	998	50	"
09 / 0000	16.2	104.5	995	60	“
09 / 0600	16.7	105.3	991	65	hurricane
09 / 1200	17.3	105.9	991	65	“
09 / 1800	17.9	106.6	991	65	“
10 / 0000	18.5	107.3	994	60	tropical storm
10 / 0600	19.1	107.9	1000	50	“
10 / 1200	19.8	108.2	1002	45	“
10 / 1800	20.5	108.1	1005	35	low
11 / 0000	21.0	107.9	1007	25	“
11 / 0600	21.3	107.5	1009	20	“
11 / 1200	21.5	107.2	1010	15	“
11 / 1800	21.7	107.2	1010	15	“
12 / 0000					dissipated
09 / 0600	16.7	105.3	991	65	maximum winds & minimum pressure

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	90	222
Medium (40%-60%)	60	186
High (>60%)	30	108

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Hurricane Barbara, 8–10 June 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**15.6**	**21.0**	**37.4**	66.9				
OCD5	24.2	57.3	95.5	114.8				
Forecasts	8	6	4	2				
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Hurricane Barbara, 8–10 June 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	17.8	21.5	35.9	82.7				
OCD5	26.8	64.6	105.2	124.8				
GFSI	31.6	45.2	56.7	**82.4**				
HWFI	33.8	44.0	77.4	197.6				
HMNI	23.8	27.3	45.9	133.2				
HFAI	29.1	45.9	91.7	231.5				
HFBI	26.9	37.1	59.4	217.9				
EMXI	21.7	24.6	**11.3**	**8.2**				
NVGI	38.3	62.7	79.2	**69.1**				
CMCI	33.4	77.3	170.8	356.0				
TVCE	21.7	26.8	36.5	143.3				
TVCX	21.3	25.5	**32.0**	121.1				
GFEX	20.1	29.7	**22.4**	**45.3**				
TVDG	21.0	24.2	**29.6**	115.6				
HCCA	18.8	**16.0**	**14.7**	**32.1**				
FSSE	19.6	**20.5**	**19.6**	**43.5**				
AEMI	28.6	41.6	55.5	123.6				
TABS	30.8	79.6	131.8	140.3				
TABM	27.3	49.3	82.9	148.9				
TABD	31.6	57.3	82.3	103.5				
Forecasts	7	5	3	1				

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Hurricane Barbara, 8–10 June 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**3.1**	**6.7**	11.2	**10.0**				
OCD5	6.9	10.0	9.8	12.5				
Forecasts	8	6	4	2				
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Hurricane Barbara, 8–10 June 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	3.1	6.7	11.2	10.0				
OCD5	6.9	10.0	**9.8**	12.5				
HWFI	7.5	**4.7**	**7.8**	**4.5**				
HMNI	5.9	7.2	**8.0**	**8.5**				
HFAI	5.9	**3.2**	**5.5**	11.0				
HFBI	7.6	**6.3**	**9.5**	**5.0**				
DSHIP	6.8	9.8	13.5	28.5				
LGEM	7.0	10.0	**9.2**	15.5				
ICON	4.1	**5.5**	**7.5**	10.5				
IVCN	3.6	**4.0**	**6.0**	**4.5**				
IVDR	3.8	**3.2**	**4.8**	**2.0**				
GFSI	5.5	**2.8**	**5.5**	10.5				
EMXI	4.6	**6.0**	**3.8**	2.0				
HCCA	5.5	6.7	13.0	**7.0**				
FSSE	3.1	**3.2**	**4.8**	**4.0**				
Forecasts	8	6	4	2				

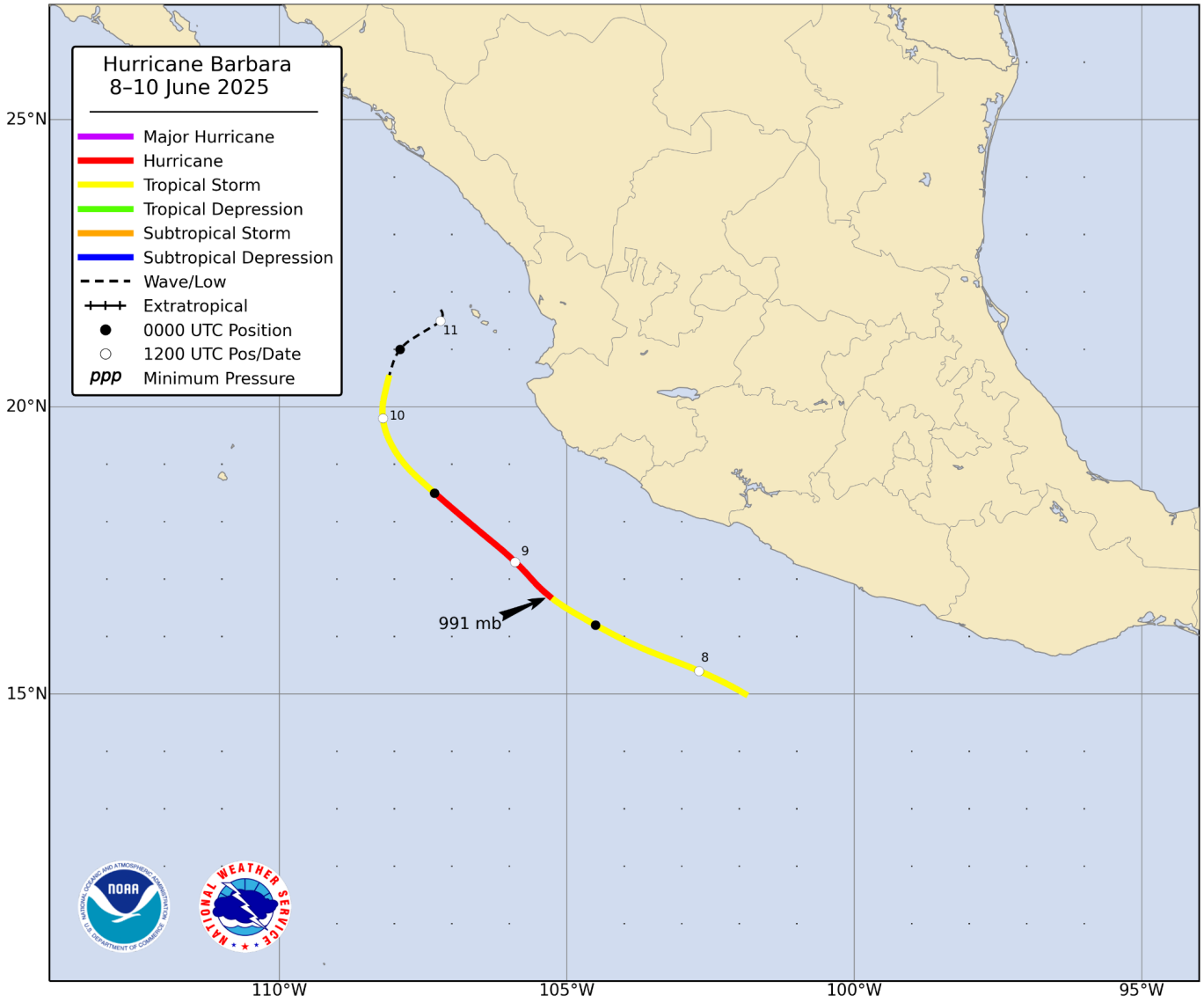


Figure 1. Best track positions for Hurricane Barbara, 8–10 June 2025.

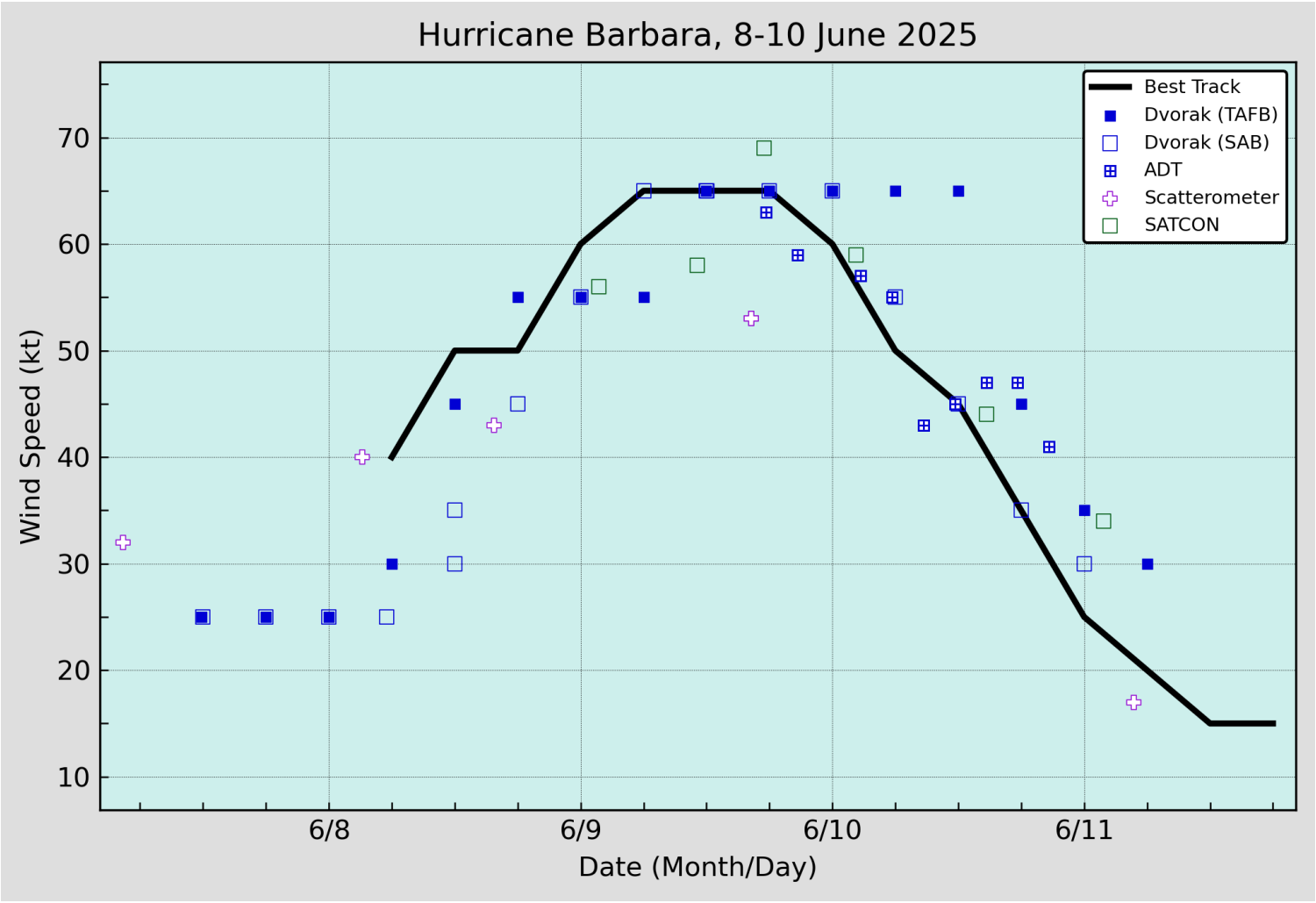


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Hurricane Barbara, 8–10 June 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

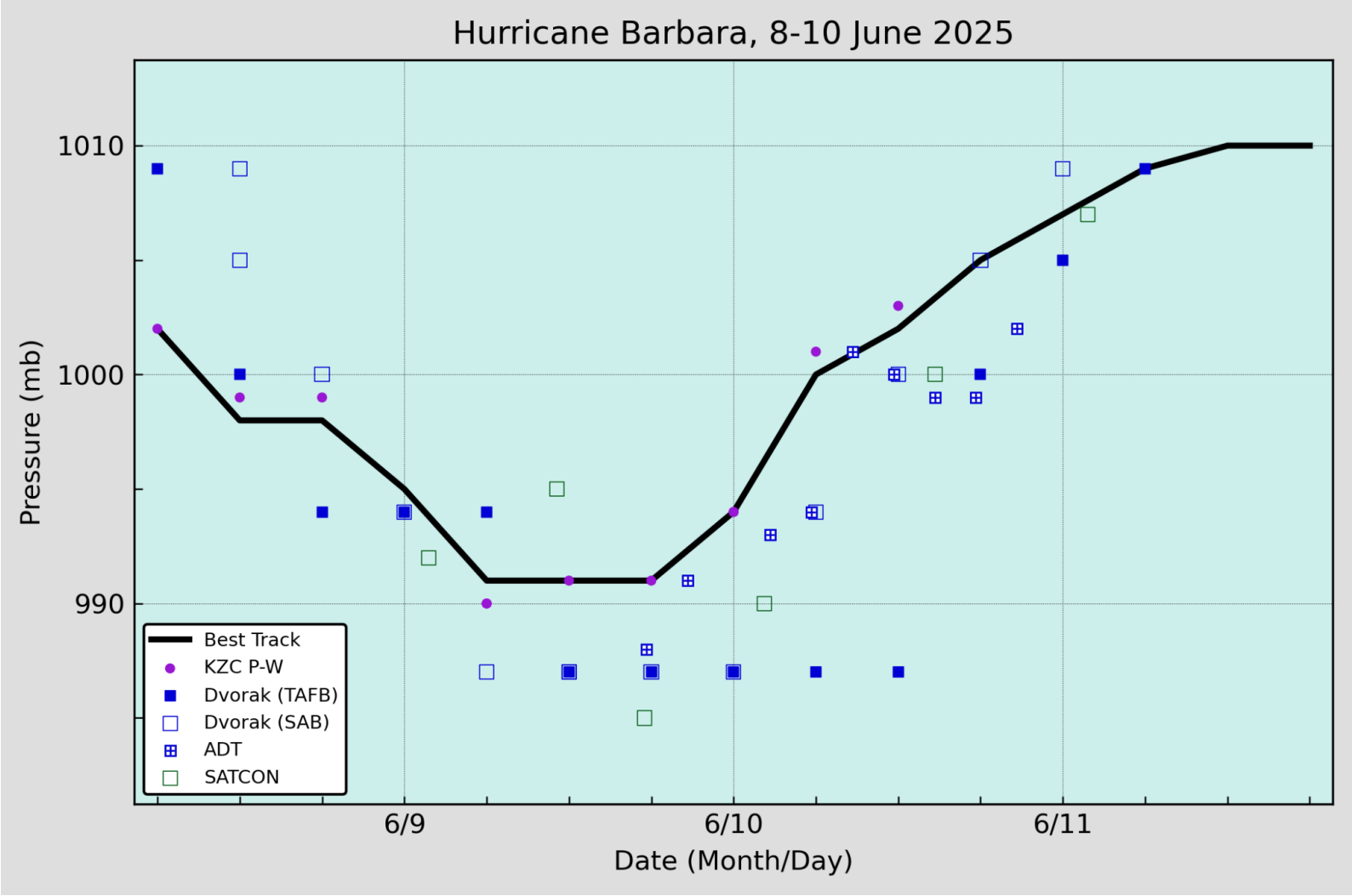


Figure 3. Selected pressure observations and best track minimum central pressure curve for Hurricane Barbara, 8–10 June 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

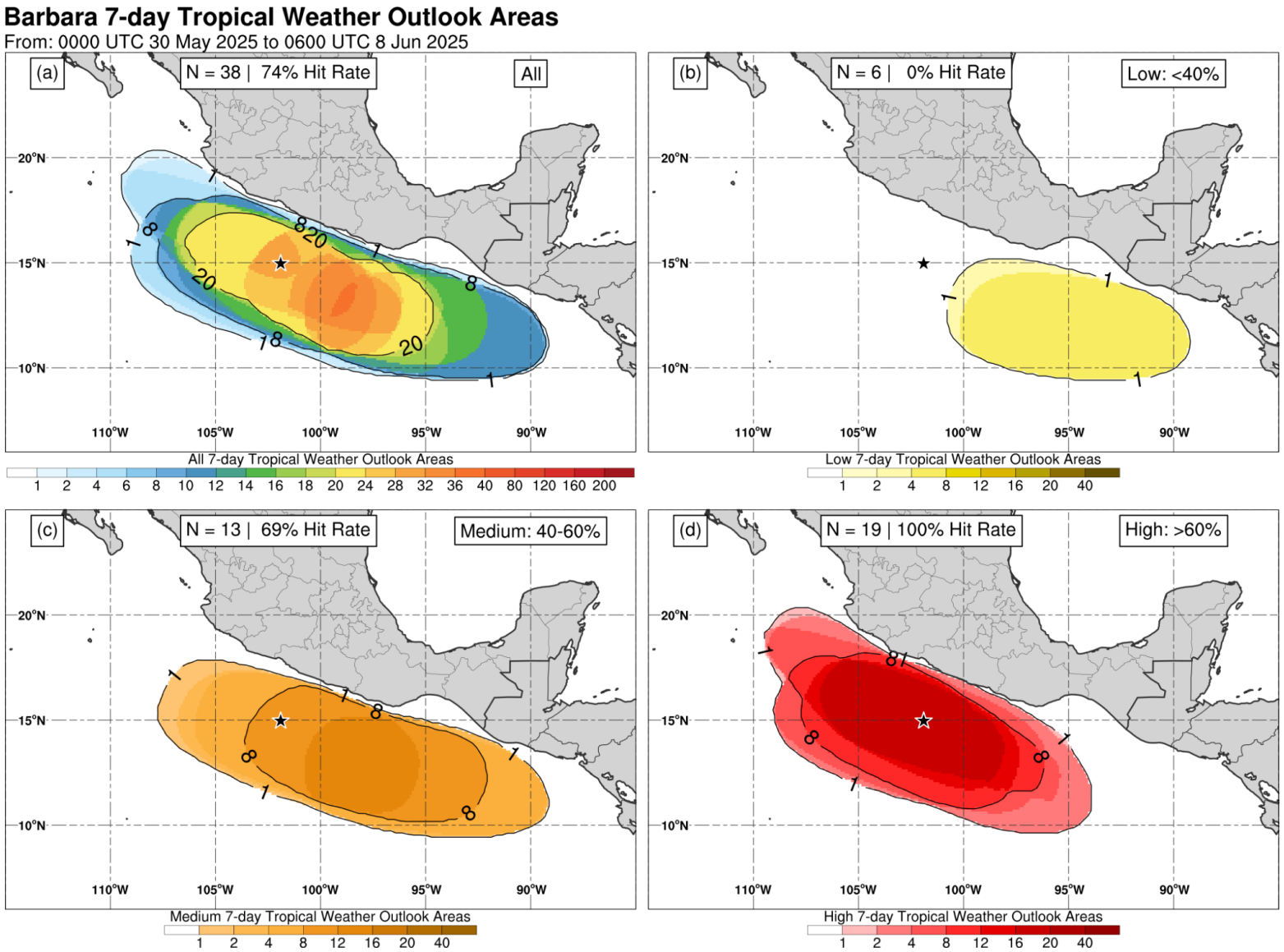


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Barbara for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.

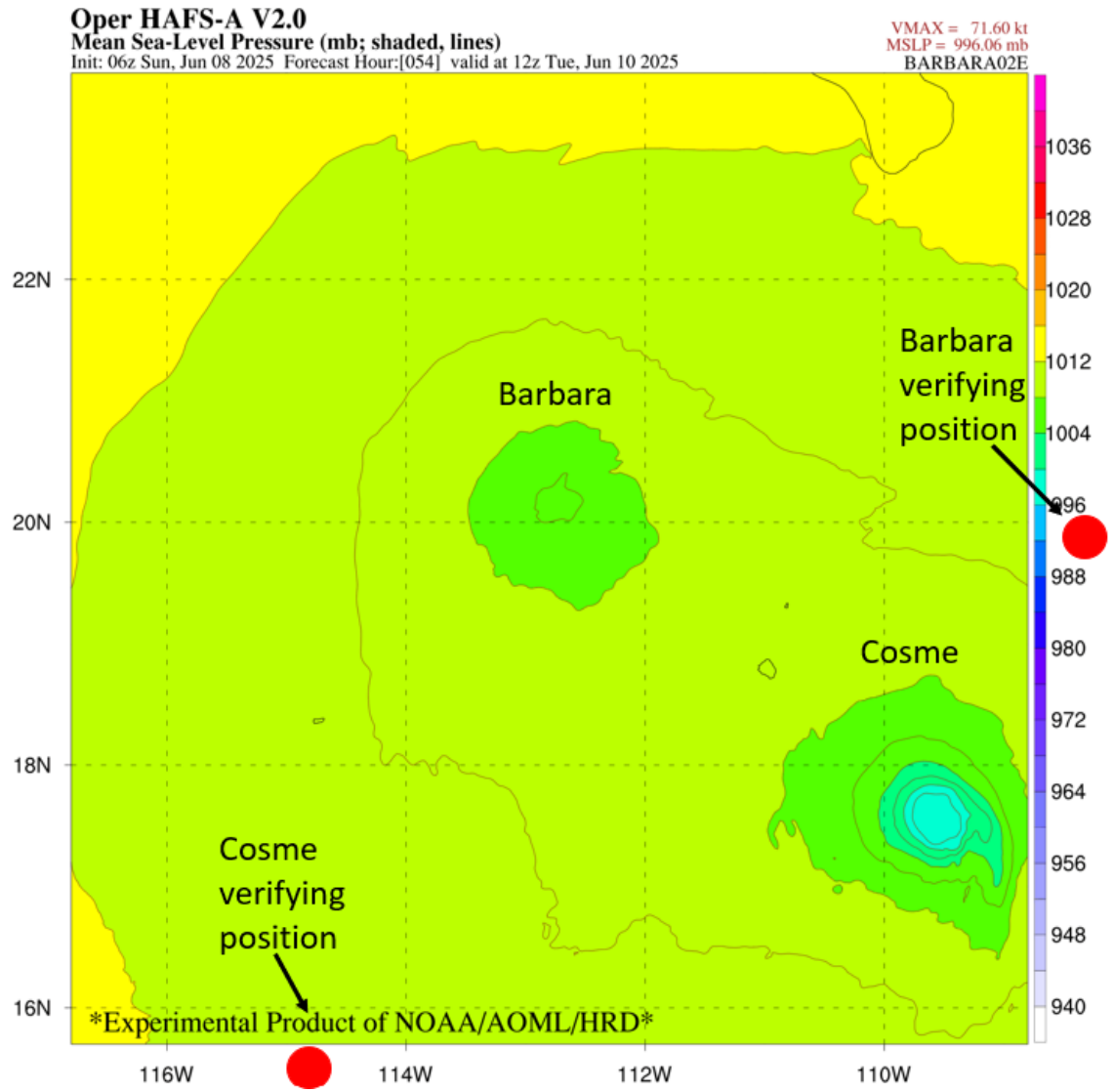


Figure 5. HAFS-A 54-h forecast initialized 8 June at 0600 UTC valid for 10 June at 1200 UTC. Mean sea-level pressure is plotted. Barbara, which had been located to the northeast of Cosme during the first day of this model run, rotated around Cosme's north side and was located to the northwest of Cosme by hour 54. The model forecast had Barbara over 200 n mi west of its verifying position (red dot), and Cosme had a similarly large error too far to the east for this HAFS-A run.

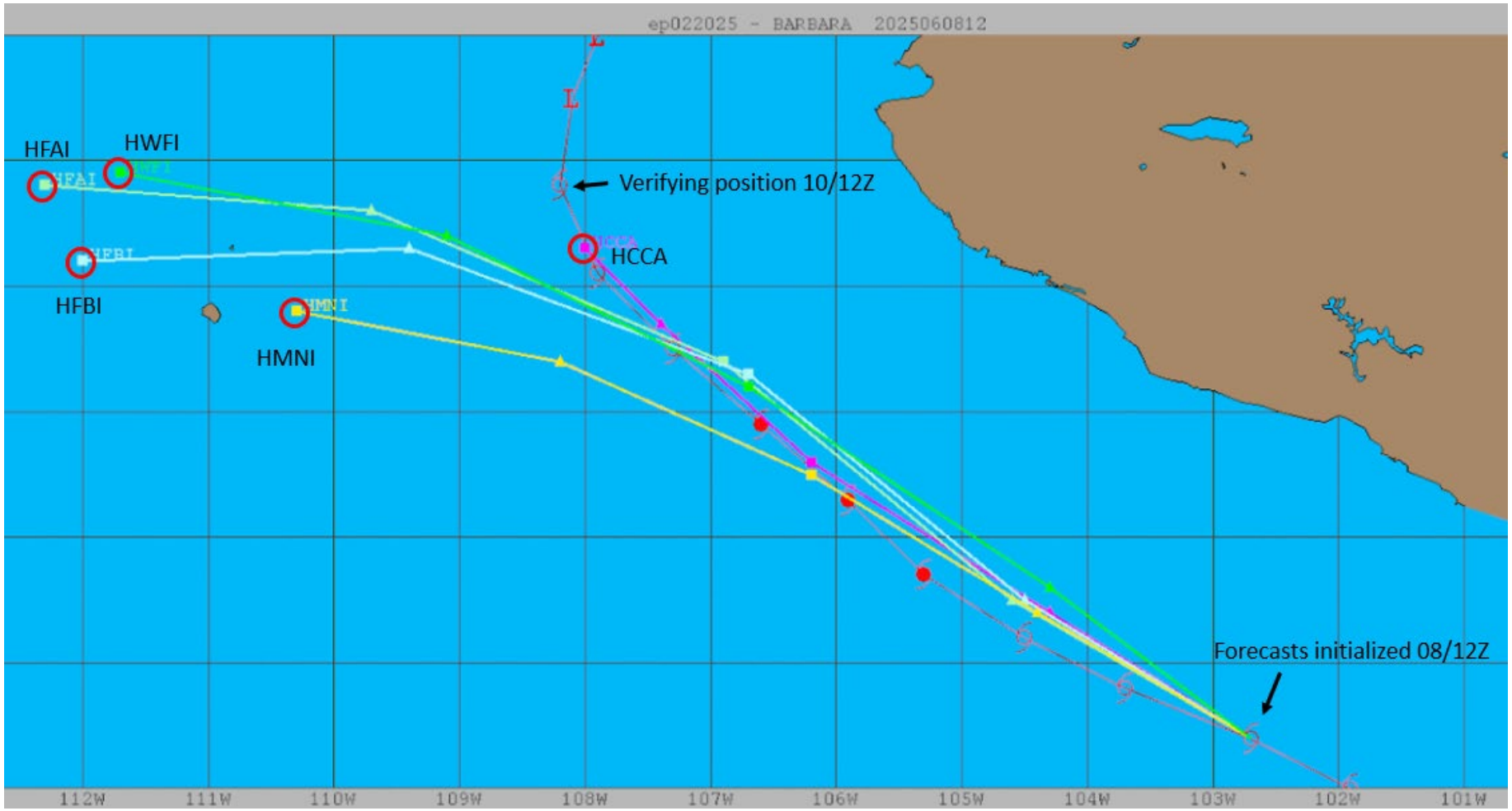


Figure 6. Best track of Hurricane Barbara (red line with red tropical cyclone symbols) plotted along with the 48-h track forecast points (red circles) for the HCCA, HFAI, HFBI, HWFI, and HMNI models with initial time of 8 June at 1200 UTC and verifying time of 10 June at 1200 UTC. The regional hurricane models turned Barbara westward by 36 h into the forecast, away from the actual track that occurred.